LAPORAN PRAKTIKUM

POSTTEST 5

ALGORITMA PEMROGRAMAN LANJUT

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PROGRAM STUDI INFORMATIKA

UNIVERSITAS MULAWARMAN

SAMARINDA

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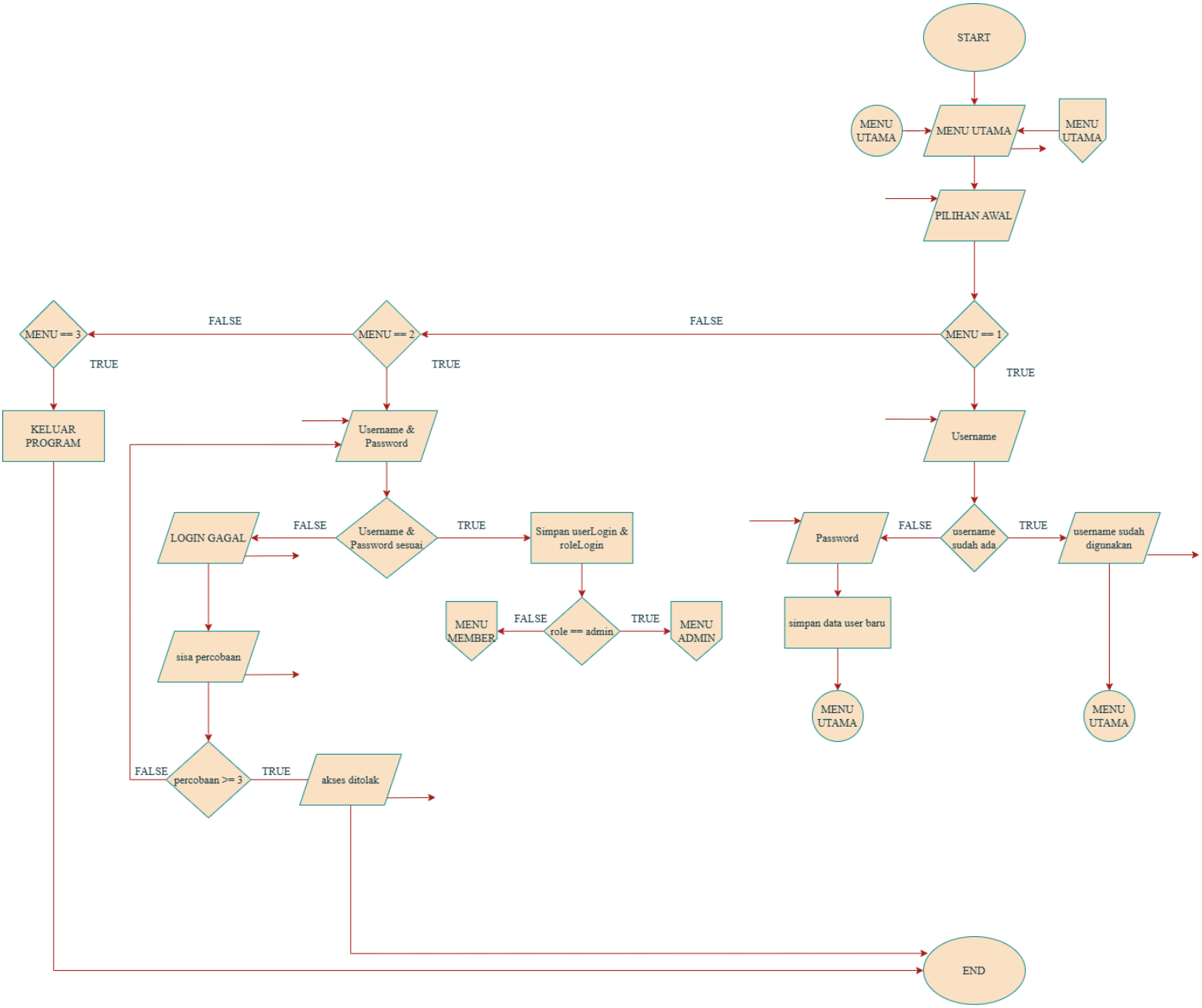
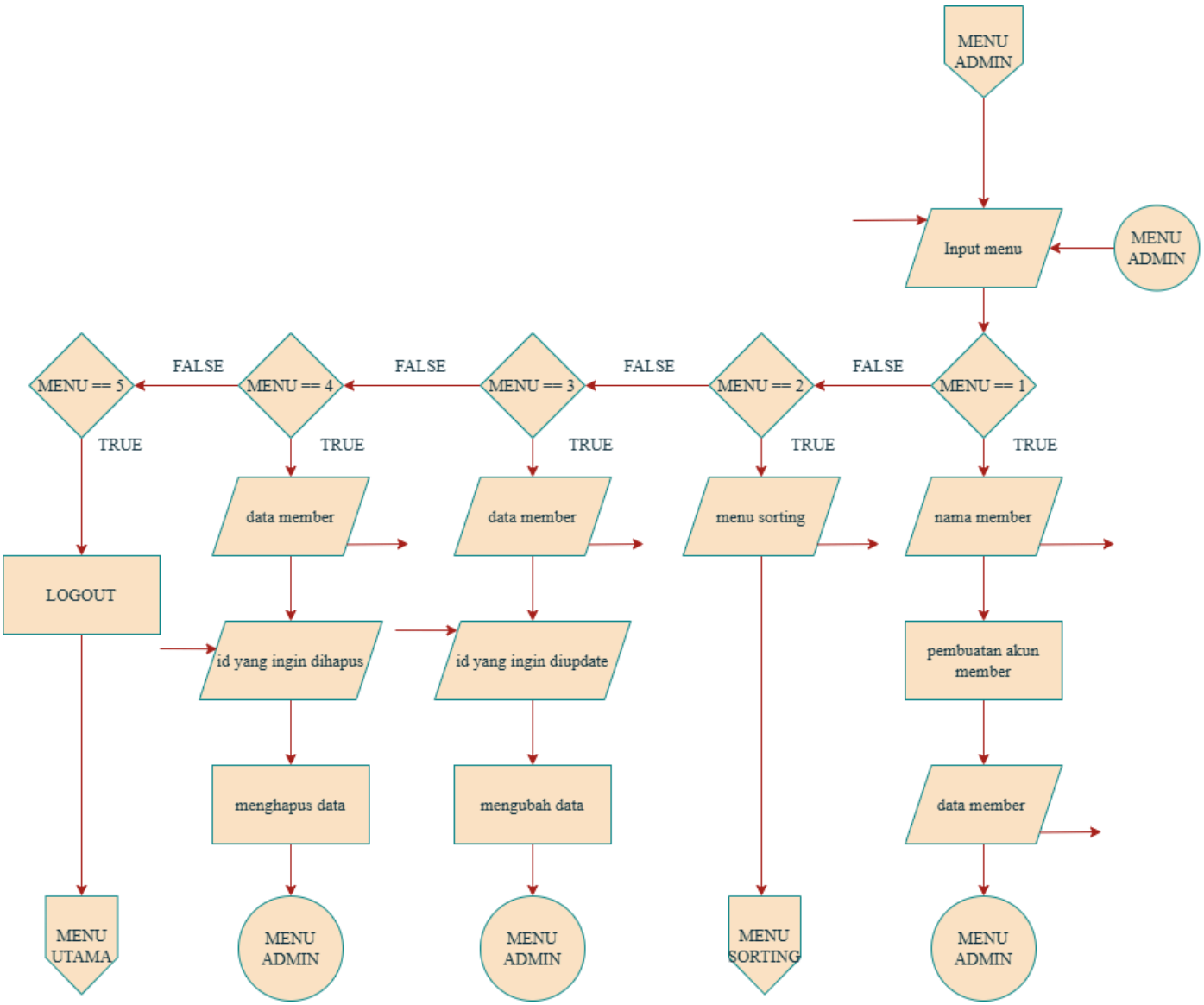
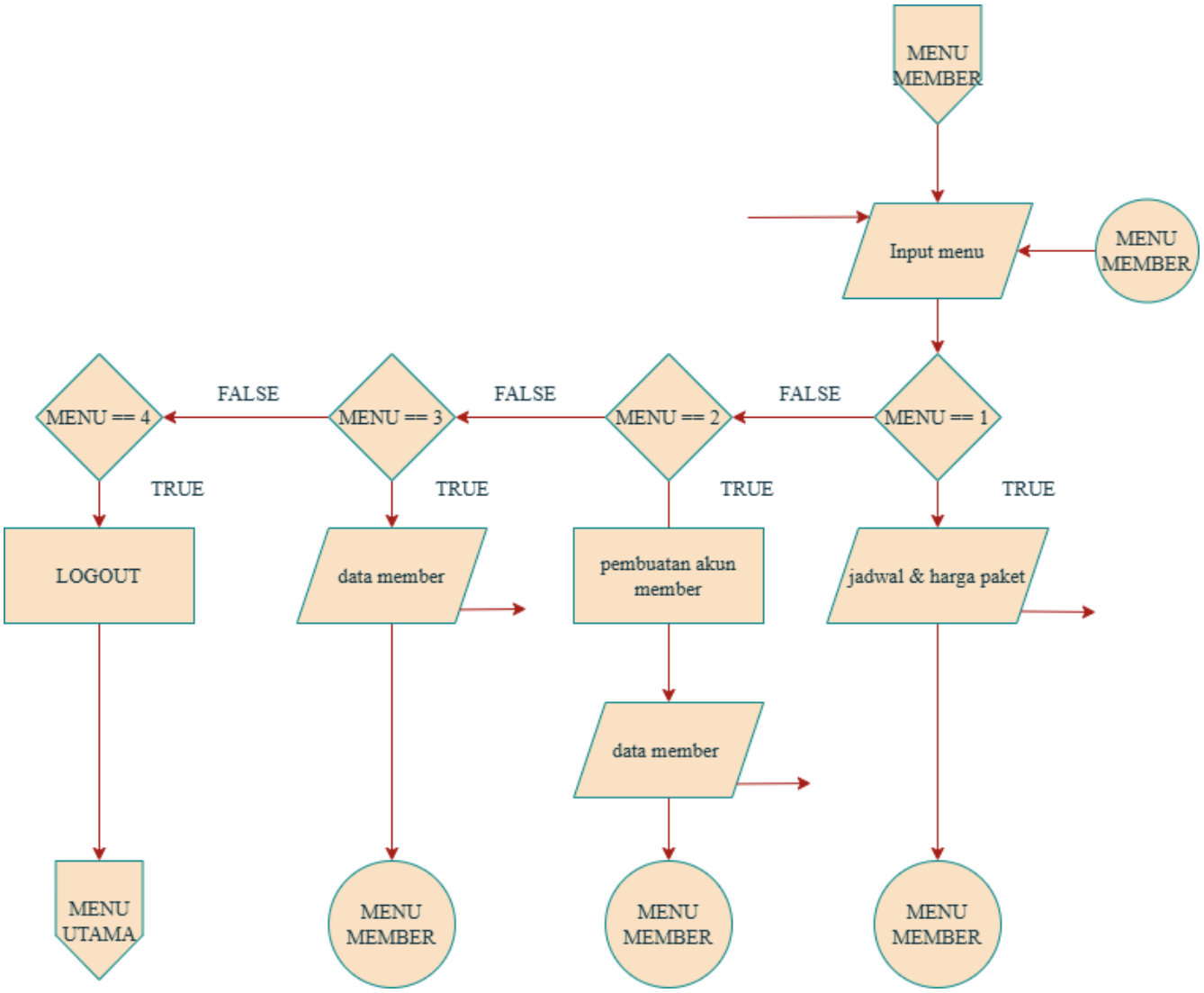
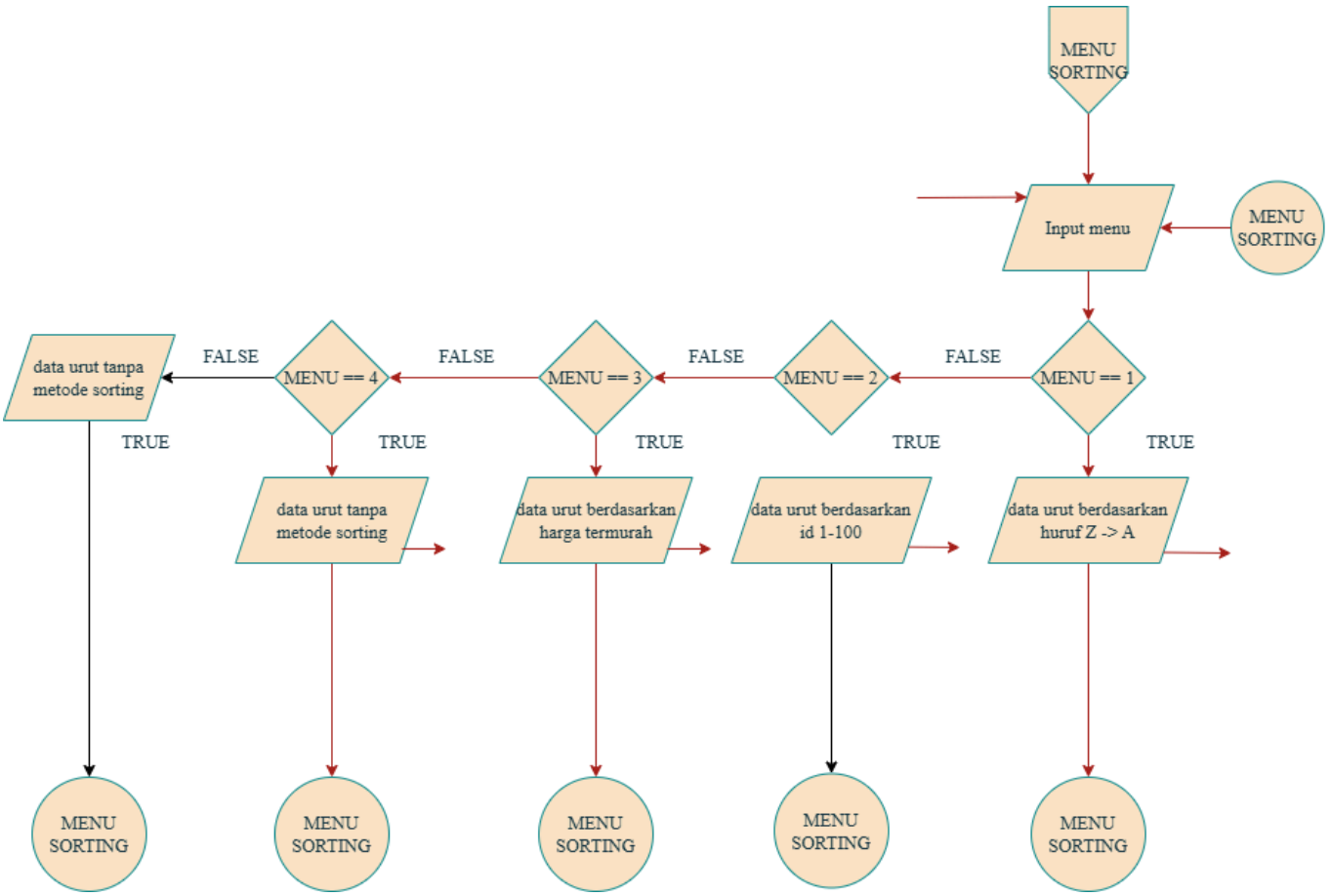
1. Flowchart


START
MENU UTAMA
MENU UTAMA
MENU UTAMA
PILIHAN AWAL
MENU == 1
FALSE
MENU == 2
FALSE
MENU == 3
TRUE
TRUE
TRUE
KELUAR PROGRAM
Username & Password
Username
LOGIN GAGAL
FALSE
Username & Password sesuai
TRUE
Simpan userLogin & roleLogin
Password
FALSE
username sudah ada
TRUE
username sudah digunakan
MENU MEMBER
FALSE
role == admin
TRUE
MENU ADMIN
simpan data user baru
sisa percobaan
MENU UTAMA
MENU UTAMA
FALSE
percobaan >= 3
TRUE
akses ditolak
END


MENU ADMIN
Input menu
MENU ADMIN
MENU == 1
FALSE
MENU == 2
FALSE
MENU == 3
FALSE
MENU == 4
FALSE
MENU == 5
TRUE
TRUE
TRUE
TRUE
TRUE
nama member
menu sorting
data member
data member
pembuatan akun member
data member
id yang ingin diupdate
id yang ingin dihapus
mengubah data
menghapus data
LOGOUT
MENU ADMIN
MENU SORTING
MENU ADMIN
MENU ADMIN
MENU UTAMA


MENU MEMBER
Input menu
MENU MEMBER
MENU == 1
FALSE
MENU == 2
FALSE
MENU == 3
FALSE
MENU == 4
TRUE
TRUE
TRUE
TRUE
jadwal & harga paket
pembuatan akun member
data member
data member
LOGOUT
MENU MEMBER
MENU MEMBER
MENU MEMBER
MENU UTAMA



1. **Flowchart Menu Utama**

- START → Program dimulai
- Tampilkan MENU UTAMA dengan 3 pilihan:
    1. Registrasi Akun Baru
    2. Login
    3. Keluar
- PILIHAN AWAL:
    1. Jika MENU == 1 (REGISTRASI):
        - Input Username
        - Cek apakah username sudah ada?
        - TRUE → Tampil "username sudah digunakan" → Kembali ke MENU UTAMA
        - FALSE → Input Password → Simpan data user baru → Kembali ke MENU UTAMA
    2. B. Jika MENU == 2 (LOGIN):

- Input Username & Password
- Cek kredensial (username & password sesuai?)
- FALSE → LOGIN GAGAL → Hitung sisa percobaan
- Jika percobaan >= 3 → AKSES DITOLAK → Program END
- Jika percobaan < 3 → Kembali input login
- TRUE → Simpan userLogin & roleLogin
- Cek role == admin?
- TRUE → Masuk ke MENU ADMIN
- FALSE → Masuk ke MENU MEMBER

3. Jika MENU == 3 (KELUAR):
    - KELUAR PROGRAM → END

2. Flowchart Menu Admin

- MENU == 1 → Buat Akun Member Baru
    1. Input nama member
    2. Proses pembuatan akun member
    3. Simpan data member → Kembali ke MENU ADMIN
- MENU == 2 → Tampilkan Seluruh Data Member
    1. Menampilkan semua data dalam array
    2. Tampil data member → Kembali ke MENU ADMIN
- MENU == 3 → Update Data Member
    1. Input id yang ingin diupdate
    2. Tampil data member yang akan diubah
    3. Proses mengubah data (paket, hari, jam, durasi)
    4. Simpan perubahan → Kembali ke MENU ADMIN
- MENU == 4 → Hapus Data Member
    1. Input id yang ingin dihapus
    2. Tampil data member yang akan dihapus
    3. Proses menghapus data (shift array)
    4. Kembali ke MENU ADMIN
- MENU == 5 → LOGOUT → Kembali ke MENU UTAMA

3. Flowchart Menu Member

- MENU == 1 → Tampilkan jadwal & harga paket → Kembali ke MENU MEMBER
- MENU == 2 → Pembuatan akun member
    1. Input data member (nama, paket, hari, jam, durasi)
    2. Simpan data member → Kembali ke MENU MEMBER
- MENU == 3 → Tampilkan data member (data diri sendiri) → Kembali ke MENU MEMBER
- MENU == 4 → LOGOUT → Kembali ke MENU UTAMA

4. Flowchart Menu Sorting

- MENU == 1 → Data urut berdasarkan huruf Z -> A (Insertion Sort - Descending) → Kembali ke MENU SORTING
- MENU == 2 → Data urut berdasarkan id 1-100 (Insertion Sort - Ascending) → Kembali ke MENU SORTING
- MENU == 3 → Data urut berdasarkan harga termurah (Quick Sort - Ascending) → Kembali ke MENU SORTING
- MENU == 4 → Data urut tanpa metode sorting (tampilan asli) → Kembali ke MENU SORTING

2. Deskripsi Singkat Program

Program ini dibuat dengan tujuan untuk memudahkan pengelolaan data member kelas padel secara terstruktur dan efisien. Program ini memungkinkan admin untuk mengelola data member (create, read, update, delete) serta member untuk mendaftar dan melihat data diri sendiri, dengan validasi paket sesi padel (SUN/MOON/STAR), pemilihan hari sesuai jadwal, dan perhitungan biaya otomatis berdasarkan durasi member.

3. Source Code

A. MENU UTAMA

```
#include <iostream>
#include <iomanip>
using namespace std;
```

```
struct user {
    string username;
    string password;
    string role;
};

struct infopaket {
    string hari;
    string jamsesi;
    int bulanMember;
    int harga;
};

struct member {
    int id;
    string nama;
    string username;
    string namapaket;
    infopaket detail;
};

#define MAX_MEMBER 100
#define MAX_user 20
#define PERTEMUAN_PER_BULAN 4

// ========== FUNGSI GLOBAL ==========
int hitungpertemuan(int bulan) {
    if (bulan <= 0) return 0;
    return PERTEMUAN_PER_BULAN + hitungpertemuan(bulan - 1);
}

string infohari(string namapaket) {
    if (namapaket == "SUN") return "Senin-Jumat";
    else if (namapaket == "MOON") return "Senin-Jumat";
    else if (namapaket == "STAR") return "Sabtu-Minggu";
    else return "-";
}

bool validasihari(string hari, string namapaket) {
    if (namapaket == "SUN" || namapaket == "MOON") {
        return (hari == "Senin" || hari == "senin" ||
                hari == "Selasa" || hari == "selasa" ||
                hari == "Rabu" || hari == "rabu" ||
                hari == "Kamis" || hari == "kamis" ||
                hari == "Jumat" || hari == "jumat");
    } else if (namapaket == "STAR") {
```

```
        return (hari == "Sabtu" || hari == "sabtu" ||
                hari == "Minggu" || hari == "minggu");
    }
    return false;
}

string inputharivalid(string namapaket) {
    string hari;
    while (true) {
        cout << "MAU PADEL DI HARI APA (" << infohari(namapaket) << "): ";
        getline(cin, hari);
        if (validasihari(hari, namapaket)) {
            if (hari.length() > 0) {
                hari[0] = toupper(hari[0]);
                for (int i = 1; i < hari.length(); i++) hari[i] = tolower(hari[i]);
            }
            return hari;
        } else {
            cout << "[!] HARI TIDAK VALID! ";
            if (namapaket == "SUN" || namapaket == "MOON")
                cout << "Paket weekday hanya bisa: Senin-Jumat" << endl;
            else
                cout << "Paket weekend hanya bisa: Sabtu&Minggu" << endl;
        }
    }
}

int hargapaket(string namapaket, int bulan) {
    int hargaPerSesi;
    if (namapaket == "SUN") hargaPerSesi = 450;
    else if (namapaket == "MOON") hargaPerSesi = 500;
    else if (namapaket == "STAR") hargaPerSesi = 650;
    else hargaPerSesi = 0;
    int totalPertemuan = hitungpertemuan(bulan);
    return hargaPerSesi * totalPertemuan;
}

int hargapaket(string namapaket) {
    if (namapaket == "SUN") return 450;
    else if (namapaket == "MOON") return 500;
    else if (namapaket == "STAR") return 650;
    else return 0;
}

void infopilihanjam(string namapaket) {
    cout << "\n--- PILIH JAM SESI (max 1 pilihan) ---" << endl;
    if (namapaket == "SUN") {
```

```
        for (int i = 1; i <= 10; i++)
            cout << i << ". " << (5+i) << ".00-" << (6+i) << ".00" << endl;
    } else if (namapaket == "MOON" || namapaket == "STAR") {
        for (int i = 1; i <= 7; i++)
            cout << i << ". " << (15+i) << ".00-" << (16+i) << ".00" << endl;
    }
}

string jamsesi(string namapaket, int pilihan) {
    if (namapaket == "SUN" && pilihan >= 1 && pilihan <= 10) {
        int start = 5 + pilihan;
        return to_string(start) + ".00-" + to_string(start + 1) + ".00";
    } else if ((namapaket == "MOON" || namapaket == "STAR") && pilihan >= 1 && pilihan <= 7)
{
        int start = 15 + pilihan;
        return to_string(start) + ".00-" + to_string(start + 1) + ".00";
    }
    return "-";
}

void tampilkaninfopaket() {
    cout << "\n=====================================================" << endl;
    cout << "              PAKET SESI DI MILO PADEL              " << endl;
    cout << "=====================================================" << endl;
    cout << "1. SUN   : weekday, jam 06.00-16.00 | Rp 450.000/sesi" << endl;
    cout << "2. MOON  : weekday, jam 16.00-23.00 | Rp 500.000/sesi" << endl;
    cout << "3. STAR  : weekend, jam 16.00-23.00 | Rp 650.000/sesi" << endl;
    cout << "=====================================================" << endl;
    cout << "[FYI] 1 bulan = 4 pertemuan (1x seminggu)" << endl;
    cout << "=====================================================" << endl;
}

// ========== FUNGSI POINTER ==========
void updateHargaDenganReference(int &harga, int bonus) {
    harga += bonus;
}

void updateNamaDenganPointer(string *namaPtr, string namaBaru) {
    if (namaPtr != nullptr) {
        *namaPtr = namaBaru;
    }
}

void tampilkanDetailMemberDenganPointer(member *mPtr) {
    if (mPtr != nullptr) {
        cout << "\n[Pointer Demo] Detail via Pointer:" << endl;
        cout << "  ID: " << mPtr->id << " | Nama: " << mPtr->nama << endl;
```

```
        cout << "  Paket: " << mPtr->namapaket << " | Harga: Rp " << mPtr->detail.harga << ".000"
<< endl;
    }
}

void demoPointerInternal(member *m, int idx) {
    cout << "\n[Pointer Info] Member #" << idx << ":" << endl;
    cout << "  Alamat memori: " << m << endl;
    cout << "  Alamat nama: " << &(m->nama) << endl;
    cout << "  Alamat harga: " << &(m->detail.harga) << endl;

    int hargaLama = m->detail.harga;
    updateHargaDenganReference(m->detail.harga, 100);
    cout << "  Harga setelah update (+100 via &): Rp " << m->detail.harga << ".000" << endl;
    m->detail.harga = hargaLama;

    string namaLama = m->nama;
    updateNamaDenganPointer(&(m->nama), namaLama + " [VIP]");
    cout << "  Nama setelah update (via *): " << m->nama << endl;
    m->nama = namaLama;

    tampilkanDetailMemberDenganPointer(m);
}

// Deklarasi fungsi dari bagian lain
void menuAdmin(member members[], int &jumlahmember);
void menuMember(member members[], int &jumlahmember, string userLogin);

int main() {
    user users[MAX_user] = {
        {"risma", "102", "member"},
        {"admin", "milopadel", "admin"}
    };
    int jumlahuser = 2;

    member members[MAX_MEMBER];
    int jumlahmember = 0;

    string inputnama, inputpw;
    string userLogin, roleLogin;

    while (true) {
        cout << "!=====================================!" << endl;
        cout << "        WELCOME TO MILO PADEL        " << endl;
        cout << "!=====================================!" << endl;

        cout << "\n--- MENU UTAMA---" << endl;
```

```
    cout << "1. Registrasi Akun Baru" << endl;
    cout << "2. Login" << endl;
    cout << "3. Keluar" << endl;
    cout << "Pilihan: ";
    int pilihanawal;
    cin >> pilihanawal;
    cin.ignore();

    if (pilihanawal == 1) {
        cout << "\n--- REGISTRASI AKUN ---" << endl;
        cout << "username: ";
        getline(cin, inputnama);

        bool usernameada = false;
        for (int i = 0; i < jumlahuser; i++) {
            if (inputnama == users[i].username) {
                usernameada = true;
                break;
            }
        }

        if (usernameada) {
            cout << "[!] username \"" << inputnama << "\" SUDAH DIGUNAKAN!" << endl;
            continue;
        }

        cout << "Password: ";
        getline(cin, inputpw);
        users[jumlahuser].username = inputnama;
        users[jumlahuser].password = inputpw;
        users[jumlahuser].role = "user";
        jumlahuser++;

        cout << "\n[+] REGIS BERHASIL! YUK LOGIN" << endl;
        cout << endl;
    }

    else if (pilihanawal == 2) {
        cout << "\n--- LOGIN user ---" << endl;

        bool loginSukses = false;

        for (int percobaan = 0; percobaan < 3; percobaan++) {
            cout << "username: ";
            getline(cin, inputnama);
            cout << "Password: ";
            getline(cin, inputpw);
```

```
        bool usnbenar = false;
        bool pwbenar = false;

        for (int i = 0; i < jumlahuser; i++) {
            if (inputnama == users[i].username) {
                usnbenar = true;
                if (inputpw == users[i].password) {
                    pwbenar = true;
                    userLogin = inputnama;
                    roleLogin = users[i].role;
                    break;
                }
            }
        }

        if (usnbenar && pwbenar) {
            loginSukses = true;
            break;
        }

        cout << "\n[!] LOGIN GAGAL" << endl;
        if (!usnbenar) {
            cout << "    -> userNAME GA ADA NIH!" << endl;
        } else {
            cout << "    -> PASSWORD SALAH!" << endl;
        }

        if (percobaan == 2) {
            cout << "\n[!] AKSES DITOLAK! GAGAL LOGIN 3x." << endl;
            return 0;
        }

        cout << "    -> Sisa percobaan: " << (2 - percobaan) << endl << endl;
    }

    if (!loginSukses) {
        return 0;
    }

    // Panggil menu sesuai role
    if (roleLogin == "admin") {
        menuAdmin(members, jumlahmember);
    } else {
        menuMember(members, jumlahmember, userLogin);
    }
```

```
        } else if (pilihanawal == 3) {
            cout << "\nTHANK YOU! see you milovers." << endl;
            break;
        } else {
            cout << "[!] PILIHAN TIDAK VALID!" << endl;
        }
    }

    return 0;
}
```

B. MENU ADMIN

```
#include <iostream>
#include <iomanip>
using namespace std;

// Struct declarations (sama seperti di main)
struct infopaket {
    string hari;
    string jamsesi;
    int bulanMember;
    int harga;
};

struct member {
    int id;
    string nama;
    string username;
    string namapaket;
    infopaket detail;
};

// Deklarasi fungsi global yang digunakan
extern int hitungpertemuan(int bulan);
extern string inputharivalid(string namapaket);
extern void infopilihanjam(string namapaket);
extern string jamsesi(string namapaket, int pilihan);
extern int hargapaket(string namapaket, int bulan);
extern int hargapaket(string namapaket);
extern void tampilkaninfopaket();
extern void updateHargaDenganReference(int &harga, int bonus);
extern void tampilkanDetailMemberDenganPointer(member *mPtr);
extern void demoPointerInternal(member *m, int idx);
```

```
#define PERTEMUAN_PER_BULAN 4
#define MAX_MEMBER 100

// ========== FUNGSI CRUD UNTUK ADMIN ==========

void buatmember(member members[], int &jumlahmember, string username) {
    for (int i = 0; i < jumlahmember; i++) {
        if (members[i].username == username) {
            cout << "\n[!] MEMBER SUDAH ADA" << endl;
            cout << "    menu 'lihat data saya' buat lihat data sendiri" << endl;
            return;
        }
    }

    if (jumlahmember < MAX_MEMBER) {
        cout << "\n--- BUAT AKUN MEMBER ---" << endl;
        member baru;
        baru.id = jumlahmember + 1;
        baru.username = username;

        cout << "Nama: ";
        getline(cin, baru.nama);

        tampilkaninfopaket();
        cout << "Paket (SUN/MOON/STAR): ";
        getline(cin, baru.namapaket);

        if (baru.namapaket != "SUN" && baru.namapaket != "MOON" && baru.namapaket !=
"STAR") {
            cout << "[!] MAAF, PAKET INI GA ADA!" << endl;
            return;
        }

        baru.detail.hari = inputharivalid(baru.namapaket);
        infopilihanjam(baru.namapaket);
        cout << "JAM BERAPA (1-" << (baru.namapaket == "SUN" ? "10" : "7") << "): ";
        int pilihanJam;
        cin >> pilihanJam;
        cin.ignore();
        baru.detail.jamsesi = jamsesi(baru.namapaket, pilihanJam);

        cout << "\nMAU JOIN BERAPA BULAN: ";
        cin >> baru.detail.bulanMember;
        cin.ignore();

        baru.detail.harga = hargapaket(baru.namapaket);
```

```
        updateHargaDenganReference(baru.detail.harga, 0);

        members[jumlahmember] = baru;
        tampilkanDetailMemberDenganPointer(&members[jumlahmember]);
        jumlahmember++;

        int totalPertemuan = hitungpertemuan(baru.detail.bulanMember);
        int total = hargapaket(baru.namapaket, baru.detail.bulanMember);

        cout << "\n[+] YEY AKUN MEMBER UDAH ADA!" << endl;
        cout << "      Paket          : " << baru.namapaket << endl;
        cout << "      Hari           : " << baru.detail.hari << endl;
        cout << "      Jam sesi       : " << baru.detail.jamsesi << endl;
        cout << "      Durasi         : " << baru.detail.bulanMember << " bulan" << endl;
        cout << "      Pertemuan/Bulan : " << PERTEMUAN_PER_BULAN << "x" << endl;
        cout << "      Total Pertemuan : " << totalPertemuan << "x" << endl;
        cout << "      Harga /Sesi     : Rp " << baru.detail.harga << ".000" << endl;
        cout << "      TOTAL BAYAR   : Rp " << total << ".000 " << endl;

    } else {
        cout << "\n[!] KAPASITAS MEMBER UDAH FULL!" << endl;
    }
}

void bacasemuadata(member members[], int jumlahmember) {
    if (jumlahmember == 0) {
        cout << "\n[!] DATA MEMBER MASIH KOSONG" << endl;
    } else {
        cout <<
"\n==========================================================================
================" << endl;
        cout << "                              DAFTAR MEMBER PADEL                              " << endl;
        cout <<
"==========================================================================
================" << endl;
        cout << left << setw(4) << "ID"
            << setw(15) << "Nama member"
            << setw(15) << "username"
            << setw(8)  << "Paket"
            << setw(10) << "Hari"
            << setw(12) << "Jam sesi"
            << setw(10) << "Bulan"
            << setw(12) << "Total" << endl;
        cout << "--------------------------------------------------------------------------------" << endl;

        for (int i = 0; i < jumlahmember; i++) {
            int totalPertemuan = hitungpertemuan(members[i].detail.bulanMember);
```

```
        int total = hargapaket(members[i].namapaket, members[i].detail.bulanMember);

        cout << left << setw(4) << members[i].id
            << setw(18) << members[i].nama
            << setw(15) << members[i].username
            << setw(8) << members[i].namapaket
            << setw(12) << members[i].detail.hari
            << setw(12) << members[i].detail.jamsesi
            << setw(10) << (to_string(members[i].detail.bulanMember) + " bln")
            << setw(12) << ("Rp " + to_string(total) + "K") << endl;
    }
    cout <<
"==========================================================================
==========" << endl;

    if (jumlahmember > 0) {
        cout << "\n[Pointer Demo] Menampilkan info member pertama via pointer:" << endl;
        demoPointerInternal(&members[0], 1);
    }
  }
}

void updatedatamember(member members[], int jumlahmember) {
  if (jumlahmember == 0) {
    cout << "\n[!] DATA MEMBER MASIH KOSONG" << endl;
  } else {
    cout << "\nDaftar Member:" << endl;
    for (int i = 0; i < jumlahmember; i++) {
        cout << "  [" << members[i].id << "] " << members[i].nama
            << " | Paket: " << members[i].namapaket << endl;
    }

    int index;
    cout << "\nID yang mau diupdate: ";
    cin >> index;
    cin.ignore();

    if (index > 0 && index <= jumlahmember) {
        int idx = index - 1;
        cout << "\n--- UPDATE DATA MEMBER ---" << endl;
        cout << "Nama: " << members[idx].nama << endl;
        cout << "Paket sekarang: " << members[idx].namapaket << endl;

        tampilkaninfopaket();
        cout << "Paket Baru (SUN/MOON/STAR): ";
        getline(cin, members[idx].namapaket);
```

```
            members[idx].detail.hari = inputharivalid(members[idx].namapaket);
            infopilihanjam(members[idx].namapaket);
            cout << "Pilih jam sesi (1-" << (members[idx].namapaket == "SUN" ? "10" : "7") << "): ";
            int pilihanJam;
            cin >> pilihanJam;
            cin.ignore();
            members[idx].detail.jamsesi = jamsesi(members[idx].namapaket, pilihanJam);

            cout << "MAU JOIN BERAPA BULAN: ";
            cin >> members[idx].detail.bulanMember;
            cin.ignore();

            members[idx].detail.harga = hargapaket(members[idx].namapaket);

            updateHargaDenganReference(members[idx].detail.harga, 0);

            int totalPertemuan = hitungpertemuan(members[idx].detail.bulanMember);
            int total = hargapaket(members[idx].namapaket, members[idx].detail.bulanMember);

            cout << "\n[+] Paket member \"" << members[idx].nama << "\" YEY UPDATE
BERHASIL!" << endl;
            cout << "    Total Pertemuan : " << totalPertemuan << "x" << endl;
            cout << "     TOTAL BAYAR   : Rp " << total << ".000 " << endl;

            cout << "\n[Pointer Demo] Member setelah update:" << endl;
            demoPointerInternal(&members[idx], members[idx].id);
        } else {
            cout << "\n[!] ID MEMBER GA ADA!" << endl;
        }
    }
}

void deletedatamember(member members[], int &jumlahmember) {
    if (jumlahmember == 0) {
        cout << "\n[!] DATA MEMBER MASIH KOSONG!" << endl;
    } else {
        cout << "\nDaftar Member:" << endl;
        for (int i = 0; i < jumlahmember; i++) {
            cout << "  [" << members[i].id << "] " << members[i].nama << endl;
        }

        int index;
        cout << "\nID yang mau dihapus: ";
        cin >> index;
        cin.ignore();

        if (index > 0 && index <= jumlahmember) {
```

```
        cout << "\n[Pointer Demo] Member yang akan dihapus:" << endl;
        demoPointerInternal(&members[index - 1], index);

        for (int i = index - 1; i < jumlahmember - 1; i++) {
            members[i] = members[i + 1];
        }
        jumlahmember--;
        cout << "\n[+] MEMBER ID INI " << index << " BERHASIL DIHAPUS!" << endl;
    } else {
        cout << "\n[!] ID MEMBER GA ADA!" << endl;
    }
  }
}

// ========== MENU ADMIN UTAMA ==========
void menuAdmin(member members[], int &jumlahmember) {
  int pilihan;
  bool menuberjalan = true;

  do {
    cout << endl;
    cout << "==========================================" << endl;
    cout << "          MENU ADMIN MILO PADEL          " << endl;
    cout << "==========================================" << endl;
    cout << "1. Buat Akun Member Baru                  " << endl;
    cout << "2. Tampilkan Semua Data                   " << endl;
    cout << "3. Update Paket Member                    " << endl;
    cout << "4. Hapus Member                           " << endl;
    cout << "5. Logout                                 " << endl;
    cout << "==========================================" << endl;
    cout << "Pilih menu (1-5): ";
    cin >> pilihan;
    cin.ignore();

    if (pilihan < 1 || pilihan > 5) {
        cout << "[!] PILIHAN TIDAK VALID! hanya 1-5." << endl;
        continue;
    }

    switch (pilihan) {
        case 1:
            buatmember(members, jumlahmember, "admin_managed");
            break;
        case 2:
            bacasemuadata(members, jumlahmember);
            break;
        case 3:
```

```
            updatedatamember(members, jumlahmember);
            break;
        case 4:
            deletedatamember(members, jumlahmember);
            break;
        case 5:
            cout << "\n[+] BERHASIL LOGOUT! kembali ke menu utama." << endl;
            menuberjalan = false;
            break;
    }
  } while (menuberjalan);
}
```

C. MENU MEMBER

```
#include <iostream>
#include <iomanip>
using namespace std;

// Struct declarations
struct infopaket {
    string hari;
    string jamsesi;
    int bulanMember;
    int harga;
};

struct member {
    int id;
    string nama;
    string username;
    string namapaket;
    infopaket detail;
};

// Deklarasi fungsi global yang digunakan
extern int hitungpertemuan(int bulan);
extern int hargapaket(string namapaket, int bulan);
extern int hargapaket(string namapaket);
extern void tampilkaninfopaket();
extern void tampilkanDetailMemberDenganPointer(member *mPtr);

#define PERTEMUAN_PER_BULAN 4

// ========== FUNGSI UNTUK MEMBER ==========
```

```
void buatmember(member members[], int &jumlahmember, string username) {
    for (int i = 0; i < jumlahmember; i++) {
        if (members[i].username == username) {
            cout << "\n[!] MEMBER SUDAH ADA" << endl;
            cout << "    menu 'lihat data saya' buat lihat data sendiri" << endl;
            return;
        }
    }

    if (jumlahmember < 100) { // MAX_MEMBER = 100
        cout << "\n--- BUAT AKUN MEMBER ---" << endl;
        member baru;
        baru.id = jumlahmember + 1;
        baru.username = username;

        cout << "Nama: ";
        getline(cin, baru.nama);

        tampilkaninfopaket();
        cout << "Paket (SUN/MOON/STAR): ";
        getline(cin, baru.namapaket);

        if (baru.namapaket != "SUN" && baru.namapaket != "MOON" && baru.namapaket !=
"STAR") {
            cout << "[!] MAAF, PAKET INI GA ADA!" << endl;
            return;
        }

        // Validasi hari untuk member
        string infohari(string namapaket) {
            if (namapaket == "SUN") return "Senin-Jumat";
            else if (namapaket == "MOON") return "Senin-Jumat";
            else if (namapaket == "STAR") return "Sabtu-Minggu";
            else return "-";
        }

        bool validasihari(string hari, string namapaket) {
            if (namapaket == "SUN" || namapaket == "MOON") {
                return (hari == "Senin" || hari == "senin" ||
                        hari == "Selasa" || hari == "selasa" ||
                        hari == "Rabu" || hari == "rabu" ||
                        hari == "Kamis" || hari == "kamis" ||
                        hari == "Jumat" || hari == "jumat");
            } else if (namapaket == "STAR") {
                return (hari == "Sabtu" || hari == "sabtu" ||
                        hari == "Minggu" || hari == "minggu");
```

```
        }
        return false;
    }

    string inputharivalid(string namapaket) {
        string hari;
        while (true) {
            cout << "MAU PADEL DI HARI APA (" << infohari(namapaket) << "): ";
            getline(cin, hari);
            if (validasihari(hari, namapaket)) {
                if (hari.length() > 0) {
                    hari[0] = toupper(hari[0]);
                    for (int i = 1; i < hari.length(); i++) hari[i] = tolower(hari[i]);
                }
                return hari;
            } else {
                cout << "[!] HARI TIDAK VALID! ";
                if (namapaket == "SUN" || namapaket == "MOON")
                    cout << "Paket weekday hanya bisa: Senin-Jumat" << endl;
                else
                    cout << "Paket weekend hanya bisa: Sabtu&Minggu" << endl;
            }
        }
    }

    void infopilihanjam(string namapaket) {
        cout << "\n--- PILIH JAM SESI (max 1 pilihan) ---" << endl;
        if (namapaket == "SUN") {
            for (int i = 1; i <= 10; i++)
                cout << i << ". " << (5+i) << ".00-" << (6+i) << ".00" << endl;
        } else if (namapaket == "MOON" || namapaket == "STAR") {
            for (int i = 1; i <= 7; i++)
                cout << i << ". " << (15+i) << ".00-" << (16+i) << ".00" << endl;
        }
    }

    string jamsesi(string namapaket, int pilihan) {
        if (namapaket == "SUN" && pilihan >= 1 && pilihan <= 10) {
            int start = 5 + pilihan;
            return to_string(start) + ".00-" + to_string(start + 1) + ".00";
        } else if ((namapaket == "MOON" || namapaket == "STAR") && pilihan >= 1 && pilihan
<= 7) {
            int start = 15 + pilihan;
            return to_string(start) + ".00-" + to_string(start + 1) + ".00";
        }
        return "-";
    }
```

```
        baru.detail.hari = inputharivalid(baru.namapaket);
        infopilihanjam(baru.namapaket);
        cout << "JAM BERAPA (1-" << (baru.namapaket == "SUN" ? "10" : "7") << "): ";
        int pilihanJam;
        cin >> pilihanJam;
        cin.ignore();
        baru.detail.jamsesi = jamsesi(baru.namapaket, pilihanJam);

        cout << "\nMAU JOIN BERAPA BULAN: ";
        cin >> baru.detail.bulanMember;
        cin.ignore();

        baru.detail.harga = hargapaket(baru.namapaket);

        members[jumlahmember] = baru;
        jumlahmember++;

        int totalPertemuan = hitungpertemuan(baru.detail.bulanMember);
        int total = hargapaket(baru.namapaket, baru.detail.bulanMember);

        cout << "\n[+] YEY AKUN MEMBER UDAH ADA!" << endl;
        cout << "      Paket          : " << baru.namapaket << endl;
        cout << "      Hari           : " << baru.detail.hari << endl;
        cout << "      Jam sesi       : " << baru.detail.jamsesi << endl;
        cout << "      Durasi         : " << baru.detail.bulanMember << " bulan" << endl;
        cout << "      Pertemuan/Bulan : " << PERTEMUAN_PER_BULAN << "x" << endl;
        cout << "      Total Pertemuan : " << totalPertemuan << "x" << endl;
        cout << "      Harga /Sesi     : Rp " << baru.detail.harga << ".000" << endl;
        cout << "      TOTAL BAYAR   : Rp " << total << ".000 " << endl;

    } else {
        cout << "\n[!] KAPASITAS MEMBER UDAH FULL!" << endl;
    }
}

void bacadatasaya(member members[], int jumlahmember, string username) {
    bool ditemukan = false;

    for (int i = 0; i < jumlahmember; i++) {
        if (members[i].username == username) {
            int totalPertemuan = hitungpertemuan(members[i].detail.bulanMember);
            int total = hargapaket(members[i].namapaket, members[i].detail.bulanMember);

            cout << "\n=====================================" << endl;
            cout << "            DATA MEMBER ANDA          " << endl;
            cout << "=====================================" << endl;
```

```
            cout << "ID              : " << members[i].id << endl;
            cout << "Nama            : " << members[i].nama << endl;
            cout << "Paket           : " << members[i].namapaket << endl;
            cout << "Hari            : " << members[i].detail.hari << endl;
            cout << "Jam sesi        : " << members[i].detail.jamsesi << endl;
            cout << "Durasi          : " << members[i].detail.bulanMember << " bulan" << endl;
            cout << "Pertemuan/Bulan : " << PERTEMUAN_PER_BULAN << "x" << endl;
            cout << "Total Pertemuan : " << totalPertemuan << "x" << endl;
            cout << "Harga /Sesi     : Rp " << members[i].detail.harga << ".000" << endl;
            cout << " TOTAL BAYAR   : Rp " << total << ".000 " << endl;
            cout << "=========================================" << endl;

            cout << "\n[Pointer Demo] Akses data via pointer:" << endl;
            tampilkanDetailMemberDenganPointer(&members[i]);

            ditemukan = true;
            break;
        }
    }

    if (!ditemukan) {
        cout << "\n[!] BELUM JOIN MEMBER!" << endl;
        cout << "    YUK buat akun dulu" << endl;
    }
}

// ========== MENU MEMBER UTAMA ==========
void menuMember(member members[], int &jumlahmember, string userLogin) {
    int pilihan;
    bool menuberjalan = true;

    do {
        cout << endl;
        cout << "==============================================" << endl;
        cout << "        MENU MEMBER MILO PADEL            " << endl;
        cout << "      Login sebagai: " << userLogin << "          " << endl;
        cout << "==============================================" << endl;
        cout << "1. Lihat Jadwal & Harga Paket              " << endl;
        cout << "2. Buat Akun Member                        " << endl;
        cout << "3. Lihat Data Saya                         " << endl;
        cout << "4. Logout                                  " << endl;
        cout << "==============================================" << endl;
        cout << "Pilih menu (1-4): ";
        cin >> pilihan;
        cin.ignore();

        if (pilihan < 1 || pilihan > 4) {
```

```
            cout << "[!] PILIHAN TIDAK VALID! hanya 1-4." << endl;
            continue;
        }

        switch (pilihan) {
            case 1:
                tampilkaninfopaket();
                break;
            case 2:
                buatmember(members, jumlahmember, userLogin);
                break;
            case 3:
                bacadatasaya(members, jumlahmember, userLogin);
                break;
            case 4:
                cout << "\n[+] BERHASIL LOGOUT! kembali ke menu utama." << endl;
                menuberjalan = false;
                break;
        }
    } while (menuberjalan);
}
```

D. MENU SORTING

```
void insertionSortNamaDescending(member arr[], int n) {
    for (int i = 1; i < n; i++) {
        member key = arr[i];
        int j = i - 1;
        while (j >= 0 && arr[j].nama < key.nama) {
            arr[j + 1] = arr[j];
            j--;
        }
        arr[j + 1] = key;
    }
}

void insertionSortIdAscending(member arr[], int n) {
    for (int i = 1; i < n; i++) {
        member key = arr[i];
        int j = i - 1;
        while (j >= 0 && arr[j].id > key.id) {
            arr[j + 1] = arr[j];
            j--;
        }
        arr[j + 1] = key;
    }
```

```
}

int partitionHarga(member arr[], int low, int high) {
    int pivot = arr[high].detail.harga;
    int i = low - 1;
    for (int j = low; j < high; j++) {
        if (arr[j].detail.harga < pivot) {
            i++;
            swap(arr[i], arr[j]);
        }
    }
    swap(arr[i + 1], arr[high]);
    return i + 1;
}

void quickSortHarga(member arr[], int low, int high) {
    if (low < high) {
        int pi = partitionHarga(arr, low, high);
        quickSortHarga(arr, low, pi - 1);
        quickSortHarga(arr, pi + 1, high);
    }
}
```

4. Hasil Output

```
!==================================!
        WELCOME TO MILO PADEL
!==================================!

--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan: 1

--- REGISTRASI AKUN ---
Username: risma
[!] Username "risma" SUDAH DIGUNAKAN!
!==================================!
        WELCOME TO MILO PADEL
!==================================!

--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan:
```

Gambar 4.1 Hasil Output

```
!=====================================!
        WELCOME TO MILO PADEL
!=====================================!

--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan: 1

--- REGISTRASI AKUN ---
Username: dapa
Password: 111

[+] REGIS BERHASIL! YUK LOGIN

!=====================================!
        WELCOME TO MILO PADEL
!=====================================!

--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan:
```

Gambar 4.2 Hasil Output

```
!=================================!
        WELCOME TO MILO PADEL
!=================================!

--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan: 2

--- LOGIN USER ---
Username: risma
Password: 102

==========================================
          login sebagai: risma (member)
          mau ngapain nih???
==========================================
1. Lihat Jadwal & Harga Paket
2. Buat Akun Member
3. Lihat Data Saya
4. Logout
==========================================
Pilih menu (1-4):
```

Gambar 4.3 Hasil Output

```
--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan: 2

--- LOGIN USER ---
Username: dapa
Password: 123

[!] LOGIN GAGAL
    -> PASSWORD SALAH!
    -> Sisa percobaan: 2

Username: 124
Password: dapa

[!] LOGIN GAGAL
    -> USERNAME GA ADA NIH!
    -> Sisa percobaan: 1

Username: dapa
Password: 124

[!] LOGIN GAGAL
    -> PASSWORD SALAH!

[!] AKSES DITOLAK! GAGAL LOGIN 3x.
PS C:\Users\lenov\Downloads\praktikum_apl\post-tes
```

Gambar 4.4 Hasil Output

```
!=====================================!
        WELCOME TO MILO PADEL
!=====================================!

--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan: 2

--- LOGIN USER ---
Username: risma
Password: 102

=============================================
          login sebagai: risma (member)
          mau ngapain nih???
=============================================
1. Lihat Jadwal & Harga Paket
2. Buat Akun Member
3. Lihat Data Saya
4. Logout
=============================================
Pilih menu (1-4): 1

=====================================================
               PAKET SESI DI MILO PADEL
=====================================================
1. SUN   : weekday, jam 06.00-16.00 | Rp 450.000/sesi
2. MOON  : weekday, jam 16.00-23.00 | Rp 500.000/sesi
3. STAR  : weekend, jam 16.00-23.00 | Rp 650.000/sesi
=====================================================
```

Gambar 4.5 Hasil Output

```
[FYI] 1 bulan = 4 pertemuan (1x seminggu)
=====================================================

=============================================
          login sebagai: risma (member)
          mau ngapain nih???
=============================================
1. Lihat Jadwal & Harga Paket
2. Buat Akun Member
3. Lihat Data Saya
4. Logout
=============================================
Pilih menu (1-4): 2

--- BUAT AKUN MEMBER ---
Nama: risma

=====================================================
                PAKET SESI DI MILO PADEL
=====================================================
1. SUN   : weekday, jam 06.00-16.00 | Rp 450.000/sesi
2. MOON  : weekday, jam 16.00-23.00 | Rp 500.000/sesi
3. STAR  : weekend, jam 16.00-23.00 | Rp 650.000/sesi
=====================================================
[FYI] 1 bulan = 4 pertemuan (1x seminggu)
=====================================================
Paket (SUN/MOON/STAR): sun
[!] MAAF, PAKET INI GA ADA!
```

Gambar 4.6 Hasil Output

```
=============================================
          login sebagai: risma (member)
          mau ngapain nih???
=============================================
1. Lihat Jadwal & Harga Paket
2. Buat Akun Member
3. Lihat Data Saya
4. Logout
=============================================
Pilih menu (1-4): 2

--- BUAT AKUN MEMBER ---
Nama: risma

=====================================================
                PAKET SESI DI MILO PADEL
=====================================================
1. SUN   : weekday, jam 06.00-16.00 | Rp 450.000/sesi
2. MOON  : weekday, jam 16.00-23.00 | Rp 500.000/sesi
3. STAR  : weekend, jam 16.00-23.00 | Rp 650.000/sesi
=====================================================
[FYI] 1 bulan = 4 pertemuan (1x seminggu)
=====================================================
Paket (SUN/MOON/STAR): SUN
MAU PADEL DI HARI APA (Senin-Jumat): sabtu
[!] HARI TIDAK VALID! Paket weekday hanya bisa: Senin-Jumat
MAU PADEL DI HARI APA (Senin-Jumat): kamis
```

Gambar 4.7 Hasil Output

```
--- PILIH JAM SESI (max 1 pilihan) ---
1. 06.00 - 07.00
2. 07.00 - 08.00
3. 08.00 - 09.00
4. 09.00 - 10.00
5. 10.00 - 11.00
6. 11.00 - 12.00
7. 12.00 - 13.00
8. 13.00 - 14.00
9. 14.00 - 15.00
10. 15.00 - 16.00
JAM BERAPA (1-10): 11

MAU JOIN BERAPA BULAN: 2

[+] YEY AKUN MEMBER UDAH ADA!
      Paket           : SUN
      Hari            : Kamis
      Jam sesi        : -
      Durasi          : 2 bulan
      Pertemuan/Bulan : 4x
      Total Pertemuan : 8x
      Harga /Sesi     : Rp 450.000
      TOTAL BAYAR   : Rp 3600.000

==========================================
          login sebagai: risma (member)
          mau ngapain nih???
==========================================
1. Lihat Jadwal & Harga Paket
2. Buat Akun Member
3. Lihat Data Saya
4. Logout
==========================================
Pilih menu (1-4):
```

Gambar 4.8 Hasil Output

```
Pilih menu (1-4): 3

=====================================
              DATA MEMBER ANDA
========================================
ID               : 1
Nama             : risma
Paket            : SUN
Hari             : Kamis
Jam sesi         : -
Durasi           : 2 bulan
Pertemuan/Bulan  : 4x
Total Pertemuan  : 8x
Harga /Sesi      : Rp 450.000
 TOTAL BAYAR    : Rp 3600.000
========================================

=============================================
          login sebagai: risma (member)
          mau ngapain nih???
=============================================
1. Lihat Jadwal & Harga Paket
2. Buat Akun Member
3. Lihat Data Saya
4. Logout
=============================================
Pilih menu (1-4): 4

[+] BERHASIL LOGOUT! kembali ke menu utama.
!===================================!
        WELCOME TO MILO PADEL
!===================================!

--- MENU UTAMA---
1. Registrasi Akun Baru
```

Gambar 4.9 Hasil Output

```
=============================================
          login sebagai: risma (member)
          mau ngapain nih???
=============================================
1. Lihat Jadwal & Harga Paket
2. Buat Akun Member
3. Lihat Data Saya
4. Logout
=============================================
Pilih menu (1-4): 4

[+] BERHASIL LOGOUT! kembali ke menu utama.
!===================================!
        WELCOME TO MILO PADEL
!===================================!

--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan: 3

THANK YOU! see you milovers.
PS C:\Users\lenov\Downloads\praktikum_apl\post-t
```

Gambar 4.10 Hasil Output

```
!=====================================!
        WELCOME TO MILO PADEL
!=====================================!

--- MENU UTAMA---
1. Registrasi Akun Baru
2. Login
3. Keluar
Pilihan: 2

--- LOGIN USER ---
Username: admin
Password: milopadel

=============================================
          login sebagai: admin (admin)
          mau ngapain nih???
=============================================
1. Buat Akun Member Baru
2. Tampilkan Semua Data
3. Update Paket Member
4. Hapus Member
5. Logout
========================================
Pilih menu (1-5):
```

Gambar 4.11 Hasil Output

```
===================
  1. Nama (Z -> A)
  2. ID (1 -> 100)
  3. Harga (Termurah)
  4. Tanpa Sorting
===================
Pilihan: 1

Data diurutkan: Nama (Z -> A)

==================================================================================
                                DAFTAR MEMBER PADEL
==================================================================================
ID  Nama member    username       Paket   Hari      Jam sesi    Bulan     Total
----------------------------------------------------------------------------
1   mayy             admin_managed  STAR    Sabtu       18.00-19.00 5 bln     Rp 13000K
==============================================================================

Menampilkan info member pertama via pointer:

[Pointer Info] Member #1:
  Alamat memori: 0x96245f69e0
  Alamat nama: 0x96245f69e8
  Alamat harga: 0x96245f6a8c
  Harga setelah update (+100 via &): Rp 750.000
  Nama setelah update (via *): mayy [VIP]

Detail via Pointer:
  ID: 1 | Nama: mayy
  Paket: STAR | Harga: Rp 650.000
```

Gambar 4.12 Hasil Output

```
  1. Nama (Z -> A)
  2. ID (1 -> 100)
  3. Harga (Termurah)
  4. Tanpa Sorting
===================
Pilihan: 2

Data diurutkan: ID (1 -> 100)

==================================================================================
                                DAFTAR MEMBER PADEL
==================================================================================
ID  Nama member    username       Paket   Hari      Jam sesi    Bulan     Total
----------------------------------------------------------------------------------
1   mayy              admin_managed  STAR    Sabtu       18.00-19.00 5 bln     Rp 13000K
==================================================================================

Menampilkan info member pertama via pointer:

[Pointer Info] Member #1:
  Alamat memori: 0x96245f69e0
  Alamat nama: 0x96245f69e8
  Alamat harga: 0x96245f6a8c
  Harga setelah update (+100 via &): Rp 750.000
  Nama setelah update (via *): mayy [VIP]

Detail via Pointer:
  ID: 1 | Nama: mayy
  Paket: STAR | Harga: Rp 650.000
```

Gambar 4.13 Hasil Output

```
  1. Nama (Z -> A)
  2. ID (1 -> 100)
  3. Harga (Termurah)
  4. Tanpa Sorting
===================
Pilihan: 3

Data diurutkan: Harga (Termurah)

==========================================================================================
                                DAFTAR MEMBER PADEL
==========================================================================================
ID  Nama member    username       Paket   Hari      Jam sesi    Bulan     Total
------------------------------------------------------------------------------------
1   mayy             admin_managed  STAR    Sabtu       18.00-19.00 5 bln     Rp 13000K
==========================================================================================

Menampilkan info member pertama via pointer:

[Pointer Info] Member #1:
  Alamat memori: 0x96245f69e0
  Alamat nama: 0x96245f69e8
  Alamat harga: 0x96245f6a8c
  Harga setelah update (+100 via &): Rp 750.000
  Nama setelah update (via *): mayy [VIP]

Detail via Pointer:
  ID: 1 | Nama: mayy
  Paket: STAR | Harga: Rp 650.000
```

Gambar 4.14 Hasil Output

```
  1. Nama (Z -> A)
  2. ID (1 -> 100)
  3. Harga (Termurah)
  4. Tanpa Sorting
====================
Pilihan: 4

Menampilkan data tanpa sorting

==================================================================================
                                DAFTAR MEMBER PADEL
==================================================================================
ID  Nama member    username       Paket   Hari      Jam sesi    Bulan     Total
----------------------------------------------------------------------------------
1   mayy             admin_managed  STAR    Sabtu       18.00-19.00 5 bln     Rp 13000K
==================================================================================

Menampilkan info member pertama via pointer:

[Pointer Info] Member #1:
  Alamat memori: 0x96245f69e0
  Alamat nama: 0x96245f69e8
  Alamat harga: 0x96245f6a8c
  Harga setelah update (+100 via &): Rp 750.000
  Nama setelah update (via *): mayy [VIP]

Detail via Pointer:
  ID: 1 | Nama: mayy
  Paket: STAR | Harga: Rp 650.000
```

Gambar 4.15 Hasil Output

```
  1. Nama (Z -> A)
  2. ID (1 -> 100)
  3. Harga (Termurah)
  4. Tanpa Sorting
===================
Pilihan: 5

Menampilkan data tanpa sorting

==========================================================================================
                                DAFTAR MEMBER PADEL
==========================================================================================
ID  Nama member    username       Paket   Hari      Jam sesi    Bulan     Total
------------------------------------------------------------------------------------
1   mayy             admin_managed  STAR    Sabtu       18.00-19.00 5 bln     Rp 13000K
==========================================================================================

Menampilkan info member pertama via pointer:

[Pointer Info] Member #1:
  Alamat memori: 0x96245f69e0
  Alamat nama: 0x96245f69e8
  Alamat harga: 0x96245f6a8c
  Harga setelah update (+100 via &): Rp 750.000
  Nama setelah update (via *): mayy [VIP]

Detail via Pointer:
  ID: 1 | Nama: mayy
  Paket: STAR | Harga: Rp 650.000
```

Gambar 4.16 Hasil Output

5. Langkah-langkah GIT

5.1 GIT Add

Git Add adalah perintah yang digunakan untuk menambahkan semua file yang ada di folder tersebut ke staging area. Titik (.) berarti semua file yang berubah atau baru dibuat akan disiapkan untuk disimpan.

```
PS C:\Users\lenov\Downloads\praktikum_apl> git add .
```

Gambar 4.17 Git Add

5.2 GIT Commit

Git Commit adalah Git Commit adalah menyimpan perubahan dari staging area ke local repository. -m "finish pt 5" adalah pesan commit yang menjelaskan perubahan ini untuk "finish pt 5" (pertemuan 5).

- 4 files changed, 956 insertions(+)
- 4 files changed → Ada 4 file yang diubah/ditambahkan
- 956 insertions(+) → Sekitar 956 baris kode baru ditambahkan
- 0 deletion(-) → Tidak ada baris kode yang dihapus.

Detail perubahan:

1. create mode 100644 kelas/pertemuan-5/pertemuan-5.cpp
2. → File pertemuan-5.cpp baru dibuat di folder kelas/pertemuan-5
3. create mode 100644 kelas/pertemuan-5/pertemuan-5.exe
4. → File executable pertemuan-5.exe baru dibuat
5. create mode 100644 post-test/post-test-apl-5/2509106102-RISMAYANTY-PT-5.cpp
6. → File post-test PT-5.cpp baru dibuat
7. create mode 100644 post-test/post-test-apl-5/2509106102-RISMAYANTY-PT-5.exe
8. → File executable PT-5.exe baru dibuat.

```
PS C:\Users\lenov\Downloads\praktikum_apl> git commit -m "finish pt5"
[main c88b90a] finish pt5
 4 files changed, 956 insertions(+)
 create mode 100644 kelas/pertemuan-5/pertemuan-5.cpp
 create mode 100644 kelas/pertemuan-5/pertemuan-5.exe
 create mode 100644 post-test/post-test-apl-5/2509106102-RISMAYANTY-PT-5.cpp
 create mode 100644 post-test/post-test-apl-5/2509106102-RISMAYANTY-PT-5.exe
```

Gambar 4.18 Git Commit

5.3 GIT Push

Git Push adalah mengunggah commit dari local repository ke remote repository (GitHub).

- Enumerating objects: 13, done→ Git menghitung 13 objek yang akan diproses
- Counting objects: 100% (13/13), done → Semua objek berhasil dihitung
- Delta compression using up to 16 threads → Git menggunakan kompresi delta dengan 16 thread untuk efisiensi
- Compressing objects: 100% (10/10), done → 10 objek berhasil dikompresi
- Writing objects: 100% (10/10), 960.93 KiB | 3.18 MiB/s, done → 10 objek (ukuran 960.93 KiB) berhasil dikirim dengan kecepatan 3.18 MiB/s
- Total 10 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0) → Total 10 objek dengan 3 delta (perbedaan), tidak ada objek yang digunakan ulang
- remote: Resolving deltas: 100% (3/3), completed with 2 local objects. → Server GitHub berhasil memproses 3 delta dengan 2 objek lokal
- remote: This repository moved. Please use the new location: → Repository dipindahkan ke lokasi baru
- To https://github.com/rismayanty/praktikum-apl.git → Berhasil push ke repository GitHub
- 92598e0..c88b90a main -> main →Berhasil push ke repository GitHub dengan commit hash dari 92598e0 ke c88b90a di branch main

```
PS C:\Users\lenov\Downloads\praktikum_apl> git push -u origin main
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 16 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (10/10), 960.93 KiB | 3.18 MiB/s, done.
Total 10 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 2 local objects.
remote: This repository moved. Please use the new location:
remote:   https://github.com/rismayanty/praktikum-apl.git
To https://github.com/rismayanty/praktikum.apl.git
   92598e0..c88b90a  main -> main
```

Gambar 4.19 Git Push